Memory Reach: GRU vs MLP under PPO on MiniGrid Memory

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Code, studies and winning checkpoints

1 Motivation

- **POMDP:** only a 7 × 7 view is seen
- **Memory** must live in a state h
- **Full BPTT** costs memory ∝ length
- **Truncation** cuts gradients to L

Core question: how far back must the gradient reach, and what does that reach cost?

250 runs · ≈ 250 GPU-hours · RTX 3060 12 GB · 16 CPUs

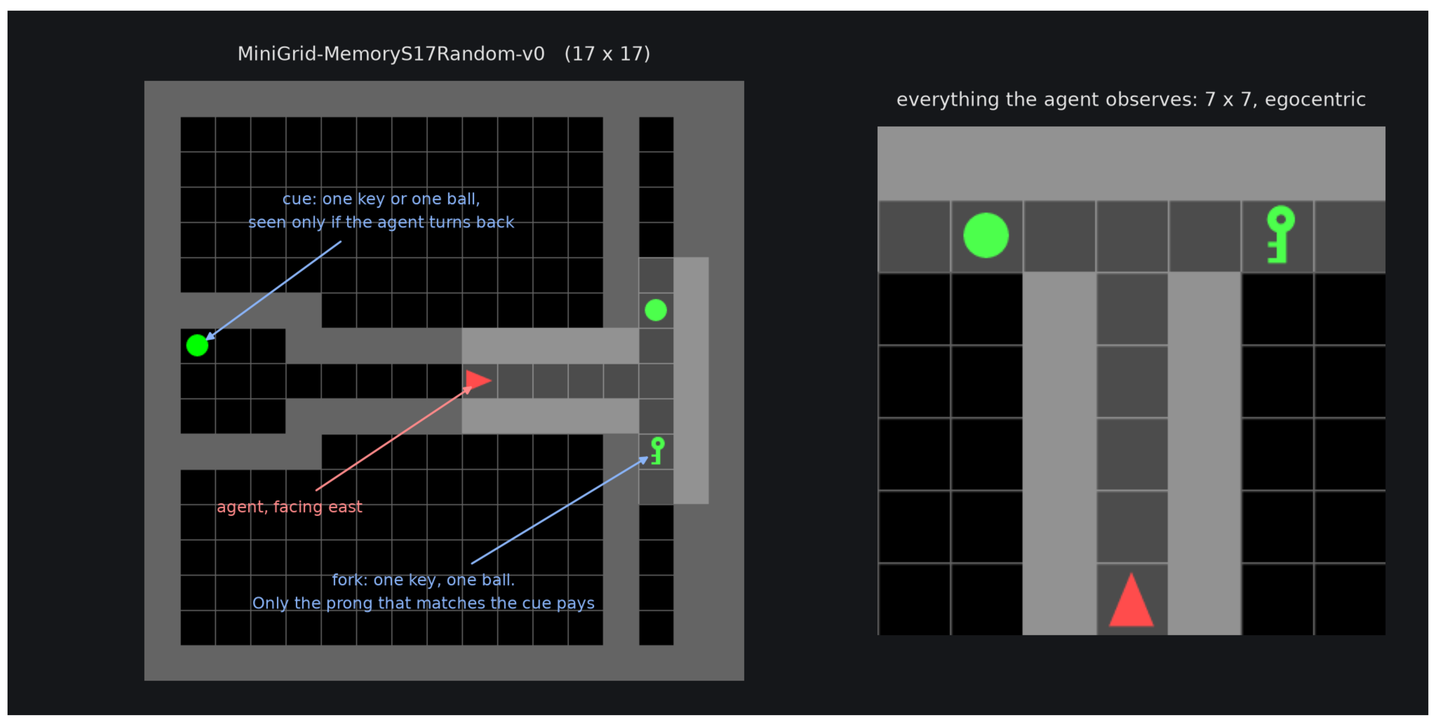
2 Problem Setting

The environment

- `MiniGrid-MemoryS17Random-v0`
- Cue (key or ball) in the west room
- Agent spawns facing away from it
- T-junction at the far east end
- Only the matching prong pays
- Reward 1 − 0.9 · steps / max_steps

The two encoders

- **MLP:** linear layers with ReLU, memory less
- **GRU:** carries h, with memory



Left: the maze. Right: everything the agent sees.

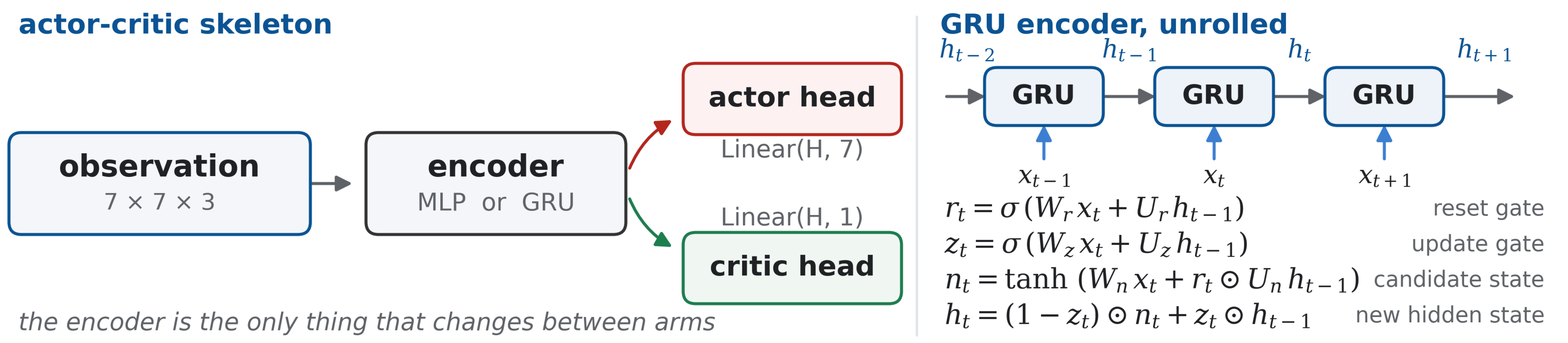
5 Key Insights

1. **Carrying h forward is not what matters.** Every GRU arm does it; only the reach of the gradient differs.
2. **What matters is the cue→decision gap.** Not the episode length. ≤ 8 steps in 29.6 % of mazes, ≤ 4 in 0%.
3. **The reach costs memory, not time.** Full BPTT is the fastest GRU arm (1.3 h); it pays in VRAM.

Recommendation. Full BPTT; L = 8 on a smaller GPU.

3 Approach

Network



PPO objective

$$\rho_t = \pi_\theta(a_t \mid s_t) \;/\; \pi_{\theta_{old}}(a_t \mid s_t)$$ probability ratio

$$\mathcal{L}^{\pi} = -\,\mathbb{E}_t\big[\min\big(\rho_t \hat{A}_t,\ \mathrm{clip}(\rho_t, 1-\varepsilon, 1+\varepsilon)\hat{A}_t\big)\big]$$ clipped policy loss

$$\mathcal{L}^{V} = \mathbb{E}_t\big[(V_\theta(s_t) - \hat{R}_t)^2\big]$$ value loss

$$\mathcal{L} = \mathcal{L}^{\pi} + c_v\,\mathcal{L}^{V} - c_e\,\mathcal{H}[\pi_\theta]$$ what is backpropagated

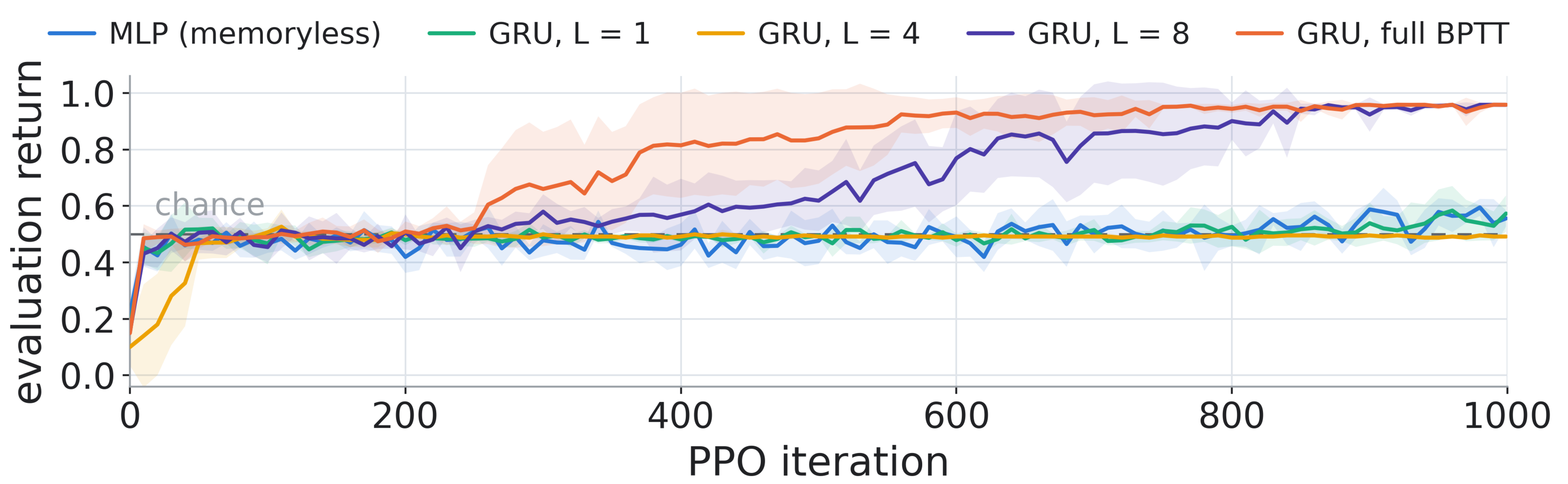
Hyperparameter Optimization

- **Surrogate model:** Optuna TPE sampler, seeded at 42
- **Budget:** 50 trials per arm × 5 arms = 250 training runs
- **Seeds per trial:** 5, each 1000 iterations
- **Objective:** mean − std of return over the 5 seeds

4 Results

arm	return	success rate	steps	minibatch	time
MLP (memoryless)	0.556 ± 0.029	0.592 ± 0.027	24.3 ± 2.3	1024	1.3 h
GRU, L = 1	0.573 ± 0.064	0.588 ± 0.068	9.8 ± 4.1	4096	2.6 h
GRU, L = 4	0.492 ± 0.001	0.500 ± 0.000	7.0 ± 0.0	4096	1.6 h
GRU, L = 8	0.958 ± 0.004	1.000 ± 0.000	16.8 ± 1.4	4096	2.0 h
GRU, full BPTT	0.958 ± 0.002	1.000 ± 0.000	16.8 ± 0.9	512	1.3 h

Winning trial per study; mean ± std across the five seeds.



6 Limitations & Future Work

- One task, one architecture, one algorithm
- **GRU is slow:** the recurrence runs step by step
- Next: **LSTM** for a bigger environment
- Next: **transformer** for a faster training time

References. Hausknecht & Stone 2015 · Kapturowski et al. 2019 (R2D2) · Ni et al. 2022 · Schulman et al. 2017 (PPO) · Chevalier-Boisvert et al. 2023 (MiniGrid)